

Importance of N'

$N'(c, \delta)$ is correct than $N(c, \delta)$ because limit never talks about the value of function at c , it only requires the neighbourhood to hold true.

Criteria

Lim exists $\rightarrow$ $LHL = RHL = L$ (finite)

Continuous $\rightarrow$ $LHL = RHL = L = f(a)$